Expansion Joints

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
Single Sphere Expansion Joint; Double Sphere Expansion Joint	IS 14533 (Rubber Expansion Joints); IS 5382; IS 7466; IS 11149; IS 10188; IS 13137; IS 3400	ISO 15348 (Rubber and Plastics Hoses and Hose Assemblies — Terms and Definitions, where applicable); ISO 23936-1 (Non-metallic Materials in Oil and Gas Applications),ISO 3601-1; ISO 3601-3; ISO 3601-5; ISO 4633; ISO 6447; ISO 9631	ASTM F1123 (Nonmetallic Expansion Joints for Piping Systems, where applicable); ASTM D2000 (Rubber Products Classification System); ASTM D471 (Effect of Liquids on Rubber); ASTM D412 (Tensile Properties of Rubber and Thermoplastic Elastomers); ASTM C1115; ASTM C361; ASTM C443; ASTM C564; ASTM C864; ASTM D1414; ASTM D1869; ASTM D5193; ASTM F104; ASTM F112; ASTM F1276; ASTM F146; ASTM F152; ASTM F2524; ASTM F36; ASTM F37; ASTM F38; ASTM F433; ASTM F495; ASTM F607; ASTM F806	DIN EN 14917 (Metal Expansion Joints, where applicable); DIN ISO 1436 (Rubber Hoses and Assemblies, where rubber expansion joints apply); DIN ISO 37 (Tensile Properties of Rubber); DIN ISO 1817 (Resistance of Rubber to Liquids)	EJMA Standards (Expansion Joint Manufacturers Association); ASME Boiler and Pressure Vessel Code (Pressure Applications); API 6A (Oil & Gas Applications, where applicable)
Flanged Expansion Joint	IS 14533 (Rubber Expansion Joints); IS 5382; IS 7466; IS 11149; IS 10188; IS 13137; IS 3400	ISO 15348 (Rubber and Plastics Hose Assemblies — Terminology); ISO 23936-1 (Non-metallic Materials in Oil and Gas Applications),ISO 3601-1; ISO 3601-3; ISO 3601-5; ISO 4633; ISO 6447; ISO 9631	ASTM F1123 (Nonmetallic Expansion Joints for Piping Systems); ASTM D2000 (Rubber Products Classification System); ASTM D2240 (Durometer Hardness of Rubber); ASTM D395 (Compression Set of Rubber); ASTM C1115; ASTM C361; ASTM C443; ASTM C564; ASTM C864; ASTM D1414; ASTM D1869; ASTM D5193; ASTM F104; ASTM F112; ASTM F1276; ASTM F146; ASTM F152; ASTM F2524; ASTM F36; ASTM F37; ASTM F38; ASTM F433; ASTM F495; ASTM F607; ASTM F806	DIN EN 1092-1 (Flanges and Their Joints — Circular Flanges); DIN EN 13555 (Gasket Parameters for Flange Connections); DIN ISO 37 (Tensile Properties); DIN ISO 815-1 (Compression Set)	ASME B16.5 (Pipe Flanges and Flanged Fittings); EJMA Standards; API 6A (Oil & Gas Applications)
PTFE Lined Expansion Joint	IS 14533 (Rubber Expansion Joints); IS 14534 (PTFE-lined Equipment Applications, where applicable); IS 5382; IS 7466; IS 11149; IS 10188; IS 13137; IS 3400	ISO 12086-1 (Fluoropolymer Materials — Designation System and Basis for Specifications); ISO 15494 (Plastics Piping Systems for Industrial Applications),ISO 3601-1; ISO 3601-3; ISO 3601-5; ISO 4633; ISO 6447; ISO 9631	ASTM D4894 (PTFE Granular Molding and Ram Extrusion Materials); ASTM F1123 (Nonmetallic Expansion Joints for Piping Systems); ASTM D638 (Tensile Properties of Plastics); ASTM D792 (Density and Specific Gravity of Plastics); ASTM C1115; ASTM C361; ASTM C443; ASTM C564; ASTM C864; ASTM D1414; ASTM D1869; ASTM D5193; ASTM F104; ASTM F112; ASTM F1276; ASTM F146; ASTM F152; ASTM F2524; ASTM F36; ASTM F37; ASTM F38; ASTM F433; ASTM F495; ASTM F607; ASTM F806	DIN EN 14741 (Thermoplastics Piping Systems — PTFE Components, where applicable); DIN EN ISO 12086-1 (Fluoropolymer Materials); DIN ISO 37 (Tensile Properties); DIN ISO 1817 (Chemical Resistance)	FDA 21 CFR 177 (Food Contact PTFE); ASME B16.5 (Flange Applications); API Standards for Chemical/Petrochemical Service

Source field: Rubber and Rubber products